

Curriculum Vitae

Dr. Martín Emilio de los Rios

Academic Information

- **2025.** Assistant researcher at IATE. Córdoba. Argentina.
- **2024-2025.** Postdoctoral researcher at SISSA. Trieste. Italy.
- **2021-2024.** Postdoctoral researcher at IFT-UAM. Madrid. Spain.
- **2019-2021.** Postdoctoral researcher at ICTP-SAIFR, FAPESP. Sao Paulo. Brazil. *Scholarship FAPESP.*
- **2014-2019.** PhD in Astronomy. *Universidad Nacional de Córdoba. Argentina. Scholarship CONICET.*
 - Thesis Title: Machine Learning Techniques applied to Cosmological Problems. Defended on 2019, April 1st.
 - PhD Advisor: Dr. Professor Mariano Domínguez.
- **2009-2014.** Title of qualification awarded: 'Licenciado' in Astronomy (equivalent to master degree). Grade Point Average: 9,07 (Max. 10). *Universidad Nacional de Córdoba. Argentina. Scholarship "estímulo a la vocación científica". SeCyT*
- Graduate with honours on high-school at Jujuy Argentina in 2008.

Languages proficiency

- *Spanish:* Native.
- *English:* Fluent. London tests of English, level 3 (upper intermediate Communication). Pass with merit in December of 2007.
- *Portuguese:* Conversational.

Programming Languages

- Extended experience with the high-level languages R and Python, as well as with the `jupyter lab` workspace. It is worth to remark that I was the main developer of several machine learning algorithms for the analysis of different astronomical data, that are publicly available on my github repository.
- Experience with low-level languages as Fortran and C++.
- I have experienced on working with different collaborative and version control tools as `git`, `docker`, `singularity` and `slack`.

Refereed Publications.

1. Shedding Light on Dark Matter at the LHC with Machine Learning. **Submitted to JHEP journal**. Arganda, [de los Rios](#), Perez, Roy, Sandá Seoane & Wagner.
2. Galaxy evolution in groups: transition galaxies in the IllustrisTNG simulations. **Published in A&A Journal, Volume 698, id.A57, 15 pp.** . Levis, Coenda, Muriel, [de los Rios](#), Ragone-Figueroa, Martínez & Ruiz.
3. Reconstructing orbits of galaxies in extreme regions (ROGER). IV. Unveiling galaxy evolution patterns in OmegaWINGS clusters. **Published in A&A Journal, Volume 695, id.A258, 9 pp.**. Muriel, Pérez-Millán, [de los Rios](#), Biviano, Coenda, Martinez, Ruiz, Vulcani & Levis.
4. Enhancing the accuracy of observable distributions for galaxies classified in the Projected Phase Space Diagram. **Published in A&A Journal, Volume 695, id.A151, 7 pp.**. Martínez, [de los Rios](#), Coenda, Muriel, Ruiz, Cora & Vega-Martínez.
5. Bayesian technique to combine independently-trained Machine-Learning models applied to direct dark matter detection. **Published in JCAP Volume 2025. (arXiv: 2407.21008)**. Cerdeño, [de los Rios](#) & Perez.
6. Machine-Learning Analysis of Radiative Decays to Dark Matter at the LHC. **Published in JHEP journal 07 (2025) 014. (arXiv: 2410.13799)**. Arganda, Carena, [de los Rios](#), Perez, Rocha, Sandá Seoane & Wagner.
7. Machine-learned exclusion limits without binning. **EPJC Volume 83, article number 1158, (2023)**. Arganda, [de los Rios](#), Perez & Sandá Seoane.
8. Measuring the sterile neutrino mass in spallation source and direct detection experiments. **JHEP Volume 2023, article number 96, (2023)**. Alonso-González, Amaral, Bariego-Quintana, Cerdeño, [de los Rios](#).
9. Imposing exclusion limits on new physics with machine-learned likelihoods. **Proceedings of 41st International Conference on High Energy physics (ICHEP2022)**. Arganda, [de los Rios](#), Perez & Sandá Seoane.
10. Search for low-mass dark matter via bremsstrahlung radiation and the Migdal effect in SuperCDMS. **Phys. Rev. D 107, 112013. As part of the SuperCDMS Collaboration.**
11. First measurement of the nuclear-recoil ionization yield in silicon at 100 eV **Phys. Rev. Lett. 131, 091801. As part of the SuperCDMS Collaboration.**
12. Backsplash galaxies and their impact on galaxy evolution: a three-stage, four-type perspective. **Monthly Notices of the Royal Astronomical Society, Volume 525, Issue 2, October 2023, Pages 3048–3060**. Ruiz, Martínez, Coenda, Muriel, Cora, [de los Rios](#), Vega-Martínez.

13. Reconstructing Orbits of Galaxies in Extreme Regions (ROGER) III: comparing galaxy populations in and around X-ray galaxy clusters. **Monthly Notices of the Royal Astronomical Society, Volume 519, Issue 3, March 2023, Pages 4360–4369** Martínez, Coenda, Muriel, de los Rios & Ruiz
14. Cosmic Kite: Auto-encoding the Cosmic Microwave Background. **Monthly Notices of the Royal Astronomical Society, Volume 511, Issue 4, pp.5525-5535.** de los Rios.
15. Reconstructing Orbits of Galaxies in Extreme Regions (ROGER) II: reliability of projected phase-space in our understanding of galaxy populations **Monthly Notices of the Royal Astronomical Society, Volume 510, Issue 2, pp.1934-1944.** Coenda, de los Rios, Muriel, Cora, Martínez, Ruiz & Vega-Martínez.
16. ROGER: Reconstructing Orbits of Galaxies in Extreme Regions using machine learning techniques. **Monthly Notices of the Royal Astronomical Society, Volume 500, Issue 2, January 2021, Pages 1784–1794 (2021).** de los Rios, Martínez, Coenda, Muriel, Ruiz, Vega-Martínez & Cora.
17. Determining the dark matter distribution in simulated galaxies with deep learning. **Monthly Notices of the Royal Astronomical Society, Volume 525, Issue 4, November 2023, Pages 6015–6035.** de los Rios, Petač, Iocco, Calore & Zaldivar.
18. I. Analysis of candidates for interacting galaxy clusters. **Astronomy & Astrophysics, Volume 611, id.A78, 12 pp. (2018).** Gonzalez, de los Rios, Oio, Hernández Lang, Tagliaferro, Domínguez, Castellón, Cuevas & Valotto.
19. Faraday Rotation Measure dependence with cluster dynamics. **Monthly Notices of the Royal Astronomical Society, Volume 487, Issue 4, p.4768-4774 (2018).** Stasyszyn & de los Rios.
20. The MeSSI (Merging Systems Identification) Algorithm and catalogue. **Monthly Notices of the Royal Astronomical Society, Vol. 458, No. 1, p. 226-232 (2016).** de los Rios, Domínguez, Paz & Merchán.
21. The first galaxy cluster discovered by the Vista Variables in the Via Lactea Survey. **The Astrophysical Journal, Volume 874, Issue 1, article id. 46, 8 pp. (2019).** Baravalle, Castellón, Alonso, Díaz Tello, Damke, Valotto, Cuevas Larenas, Sánchez, de los Rios, Minniti, Domínguez, Gurovich, Barbá, Soto & Milla Castro.
22. Construcción de un catálogo de cúmulos de galaxias en proceso de colisión: métodos y resultados. **Boletín de la Asociación Argentina de Astronomía, vol.57, p.22-24. (2015).** de los Rios, Domínguez & Paz.
23. Estudio espectroscópico y morfológico de la nebulosa planetaria NGC 2918. **Boletín de la Asociación Argentina de Astronomía, vol.55, p.59-63. (2012).** Correa, de los Rios, Hobson, Marcionni, Sánchez, Santucho, Weidmann, Valotto & Ferreira.

Participation to Conferences and Workshops

- AI goes MAD² Workshop. **Main Organizer.** October 2024. Madrid, Spain.
- 8th IBS-MultiDark-ICTP workshop. **Oral Contribution.** November 2023. Madrid, Spain.
- XV CPAN Days. **Oral Contribution.** October 2023. Santander, Spain.
- Topics in Astroparticle and Underground Physics. **Oral Contribution.** August 2023. Vienna, Austria.
- 33rd Rencontres de Blois. **Oral Contribution.** May 2023. Blois, France.
- Dark Collaboration Workshop. **Oral Contribution.** November 2022. Santander, Spain.
- AI Goes MAD. **Oral Contribution.** June 2022. Madrid, Spain.
- 19th Multidark Workshop. **Oral Contribution.** May 2022. Madrid, Spain.
- 17th International Conference on Topics in Astroparticle and Underground Physics. August 2021. Online conference hosted by IFIC Valencia.
- XI FOF Meeting. April 2021. Online conference. Córdoba. Argentina.
- 3rd South American Dark Matter Workshop. December 2020. Online conference hosted by ICTP-SAIFR. Brazil.
- Workshop on New Trends in Dark Matter. December 2020. Online conference hosted by ICTP-SAIFR. Brazil.
- Latin American Workshop on Observational Cosmology. December 2020. Online conference hosted by ICTP-SAIFR. Brazil.
- 62° Annual Meeting of the Argentinian Astronomy Society. **Oral Contribution.** October 2020. Online conference. Argentina.
- First School on Data Science and Machine Learning. December 2019. Sao Paulo. Brazil.
- 2nd Latin American School on Parallel Programming for High Performance Computing. December 2019. Sao Paulo. Brazil.
- Dark Universe Workshop – Early Universe Cosmology, Baryogenesis and Dark Matter. October 2019. Sao Paulo. Brazil.
- Linea Bootcamp. September 2019. Rio de Janeiro. Brazil.
- School on High Energy Astrophysics. August 2019. Sao Paulo. Brazil.
- III Joint ICTP Trieste-ICTP-SAIFR School on observational cosmology. July-August 2019. Sao Paulo. Brazil.
- The dark side of the Universe. July 2019. Buenos Aires. Argentina.
- II Joint Trieste/SAIFR School on Particle Physics. From 22/06/2019 to 02/07/2019. ICTP-SAIFR (Sao Paulo-Brazil).
- 2nd Binational Meeting AAA-SOCHIAS. **Oral Contribution.** October 2018. La Serena. Chile.
- LAPIS School. **Poster Contribution.** February 2018. Córdoba. Argentina.
- VIII FOF Meeting. April 2018. Córdoba. Argentina.

- CMB from A to Z School. *November 2017. Cargese. France.*
- Brazilian School of Cosmology and Gravitation. **Poster Contribution.** *July 2017. Rio de Janeiro. Brazil.*
- Grav17 Workshop. **Oral Contribution.** *April 2017. Córdoba. Argentina.*
- VII FOF Meeting. **Oral Contribution.** *April 2017. Córdoba. Argentina.*
- School on Dark Matter. *July 2016. Sao Paulo. Brazil.*
- 10° International Conference on the interconnection between Particle Physics and Cosmology. **Oral Contribution.** *July 2016. Sao Paulo. Brazil.*
- GUASA Workshop. *December 2015. Antigua. Guatemala.*
- V FOF Meeting. **Oral Contribution.** *April 2015. Córdoba. Argentina*
- 57° Annual Meeting of the Argentinian Astronomy Society. **Poster Contribution.** *September 2014. Córdoba. Argentina*
- IV FOF Meeting. **Oral Contribution.** *April 2014. Córdoba. Argentina*
- 56° Annual Meeting of the Argentinian Astronomy Society. *September 2013. Tandil. Argentina.*
- First TOROS Workshop. *July 2013. Salta. Argentina.*
- Sérsic Meeting. *April 2013. Córdoba. Argentina.*
- III FOF Meeting. *April 2013. Córdoba. Argentina.*
- 55° Annual Meeting of the Argentinian Astronomy Society. **Poster Contribution.** *September 2012. Mar del Plata. Argentina.*

Academic Visits

- Università di Napoli "Federico II"; INFN, sezione di Napoli. Italy. *From 26/11/2021 to 05/12/2021.*
- Instituto de Astronomía Teórica y Experimental. Universidad Nacional de Córdoba. Córdoba, Argentina. *From 09/03/2019 to 13/03/2019.*
- Institute for Computational Cosmology. Durham University. Durham, England. *From 20/11/2017 to 20/12/2017.*

Teaching

- Teacher assistance in Astronomía General II course at FAMAF-UNC. 2013
- Teacher assistance in Astronomía General I course at FAMAF-UNC. 2014
- Teacher in Laboratorio II course FCQ-UNC. 2018
- Teacher in 'Complementos de Física Moderna' course FAMAF-UNC. 2019

Awards obtained

- Premio Universidad Nacional de Córdoba. 2014.

Science Organization and outreach

- *October 2024* Main Organizer of the AI Goes MAD² workshop. *IFT-UAM, Madrid, Spain.*
- *2021-2024.* Main Organizer of the Machine Learning, Artificial Intelligence & High Performance Computing Journal Club @ IFT-UAM.
- *June 2022* Main Organizer of the AI Goes MAD workshop. *IFT-UAM, Madrid, Spain.*
- *May 2022* Main Organizer of the 19th Multidark Workshop. *IFT-UAM, Madrid, Spain.*
- *2019-2021.* Main Organizer of the Astrophysics & Cosmology Journal Club @ ICTP-SAIFR.
- *2014-2018.* Organizer and participant of the Noche de los museos @ Universidad Nacional de Córdoba. Observatorio Astronómico de Córdoba. Program for outreach "Museum nights", volunteer at the Astronomical Observatory.

Thesis Supervision

1. Master's Thesis Co-Director

- *Title:* Dark matter or neutrinos: What are we (not yet) seeing in dark matter direct detection experiments?
- *Academic Institution:* Universidad Autónoma de Madrid, Departamento de Física Teórica.
- *Student:* Adriana Bariego Quintana.
- *Thesis Defense date:* September 2022.